

**IIRS Outreach Program**

**AI/ML for Geodata Analytics**

**Schedule & Lecture Details**

**August 03 – 14, 2026**

| Lecture No. | Date/Time                       | Topic                                    | Syllabus                                                                                                                                                                                                                                                                      | Faculty                     |
|-------------|---------------------------------|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| L1          | August 3, 2026<br><br>1600 hrs  | GIS Data analytics                       | Introduction to Data Analytics, Data Collection methods, Data Cleaning, Exploratory Data Analysis, Data Visualization tools, Statistics for Data Analysis                                                                                                                     | Dr. Harish Chandra Karnatak |
| L2          | August 4, 2026<br><br>1600 hrs  | Image Processing Methods                 | Introduction to Digital Images, Image Enhancement and Restoration, Image Transforms, Image segmentation, Object Detection                                                                                                                                                     | Dr. Poonam Seth Tiwari      |
| L3          | August 5, 2026<br><br>1600 hrs  | Geodata Models                           | Foundation Models for Geodata Processing                                                                                                                                                                                                                                      | Dr. Ashutosh Kumar Jha      |
| L4          | August 6, 2026<br><br>1600 hrs  | Concept of Data Science                  | Overview of Spatial Data Science: Approaches and Tools                                                                                                                                                                                                                        | Sh R. J. Bhandari SAC       |
| L5          | August 7, 2026<br><br>1600 hrs  | Python for Image Processing              | Basic Image Operations, Image Enhancement (Brightness and contrast adjustment, histogram equalization) , Geometric Transformations (Scaling, translation, rotation, affine transformations), Image Filtering (Smoothing, sharpening, edge detection (Sobel, Canny, Laplacian) | Mr. Ravi Bhandari           |
| L6          | August 10, 2026<br><br>1600 hrs | Machine Learning for geospatial analysis | Basics of Machine Learning/ Implementing Machine Learning algorithm for geospatial data processing                                                                                                                                                                            | Mr. Ravi Bhandari           |
| L7          | August 11, 2026                 | ANN and Deep Learning                    | Introduction to AI in Geospatial Analysis, Introduction to Artificial                                                                                                                                                                                                         | Dr. Poonam Seth Tiwari      |

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|-----|-----------------------------|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
|     | 1600 hrs                    |                                             | Neural Networks (ANNs), Deep Learning Architectures for Geospatial Analysis: CNN, RNN, Geospatial Data Analysis Tasks: Image Classification, Object Detection and Segmentation (Instance and Semantic) |                            |
| L8  | August 12, 2026<br>1600 hrs | AI/ML for Agriculture Analytics             | Case Examples for AI/ML for Agriculture Analytics                                                                                                                                                      | Dr. Anil Kumar             |
| L9  | August 13, 2026<br>1600 hrs | Advances in AI/ML for Geo-data processing   | GeoAI for Smart Governance: From Spatial Data to Intelligent Decision                                                                                                                                  | Shri. Nilay Nilesch, NESAC |
| L10 | August 14, 2026<br>1600 hrs | Generative AI and NLP for Geodata Analytics | Fundamentals of Generative AI, Large Language Models (LLMs), Natural Language Processing (NLP) for Geodata, AI-powered Geospatial Analysis                                                             | Mr. Shubhangam Garg        |